

MANTHAN SHRIRAM JOSHI

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Education

Northeastern University, Boston MA Master of Science in Cell and Gene Therapies <i>Coursework: Stem Cells and Regeneration, Cell and Gene Therapies, Immunology, Regulatory Landscapes, Advanced Drug Delivery Systems, Bioinformatics, Scientific Communication and Biotech Enterprise</i>	December 2026
Harvard Business School, Boston MA Certificate in Management Essentials	August 2025
Manipal Institute of Technology (MAHE), Manipal India Bachelor of Technology in Biotechnology Minor: Pharmaceutical Biotechnology <i>Four-year accredited engineering program. Coursework: Bioprocess Engineering, Downstream Processing, Bioreaction Engineering, Biochemistry, Genetic Engineering, Microbiology, Bioinformatics</i>	July 2023

Skills

Process & Manufacturing	Aseptic technique; process development, optimization, and yield improvement; assay development (dose-response, cytotoxicity, kill curve design); protocol and SOP authorship; experimental troubleshooting and root-cause investigation; cell expansion, banking, and cryopreservation; scientific documentation and technical reporting.
Cell Culture & Genome Engineering	Aseptic handling of mammalian cell lines (CHO) across 5+ passages at >80% viability; end-to-end CRISPR/Cas9 knock-in and knock-out workflow design; Cas9-RNP assembly with target-specific sgRNAs; CRISPRmax lipofection; drug-selection optimization using CellTiter-Glo 2.0 cytotoxicity assays.
Molecular Biology & Analytical	Bacterial transformation, colony screening, miniprep purification; restriction-enzyme digestion for construct verification; genomic DNA extraction; Phusion PCR; primer design in Benchling; Sanger sequencing and ICE indel analysis; agarose gel electrophoresis; Nanodrop quantification; recombinant protein expression and Ni-NTA purification (IMAC); SDS-PAGE; Western blotting; ELISA.
CGT Domain Knowledge	CAR-T (autologous and allogeneic platforms); iPSC differentiation; CRISPR/Cas9 genome engineering; lentiviral vector handling and titer determination; advanced drug delivery systems; cGMP and CGT regulatory frameworks (coursework).
Commercial & Business	Market research and competitive landscaping; investor deck development; grant proposal support; technical writing for non-specialist audiences; client and stakeholder communication; cross-functional project coordination.
Instrumentation & Software	Bioreactor/fermenter operation; HPLC; gas chromatography; fluorescence microscopy; flow cytometry sample preparation; FlowJo (data analysis); Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); Microsoft Excel; Microsoft Office; ICE web tool; R (basic).

Professional Experience

Northeastern University, Teaching Assistant, Cell and Gene Therapies, Boston, MA • Taught graduate-level CGT concepts in office hours and discussion sessions, translating current CAR-T and iPSC-to-T-cell literature for students from mixed scientific backgrounds. • Reviewed student project deliverables and delivered structured written feedback; prepared literature summaries on allogeneic and autologous CAR-T platforms to support classroom case discussions.	Jan 2026 – Apr 2026
Northeastern University, Graduate Research Project, Boston, MA • Maintained CHO cell cultures across 5+ passages at >80% viability; assembled and delivered Cas9-RNP complexes with 2 HPRT1-targeting sgRNAs via CRISPRmax lipofection for an end-to-end HPRT1 knockout and knock-in workflow. • Developed and optimized a 6-thioguanine selection protocol using CellTiter-Glo 2.0 dose-response across 8 serial 3-fold dilutions, graphed in GraphPad Prism, selecting 100 μ M (~7% viability) to enrich HPRT1-KO cells; confirmed by fluorescence microscopy and prepared single-cell suspensions for flow cytometry. • Extracted genomic DNA from selected pellets, amplified HPRT1 by Phusion PCR, and submitted 8 amplicons for Sanger sequencing and ICE indel analysis to confirm gene disruption. • Validated 5 plasmid constructs by restriction digest and gel electrophoresis; designed 2 primer sets in Benchling for diagnostic and knock-in confirmation amplicons.	Sept 2025 – Dec 2025
Reagene Biosciences Pvt Ltd, Associate Scientist, Bangalore, India • Developed, optimized, and validated ELISA platforms including COVID-19 diagnostics, and designed cell-based bioassays to evaluate therapeutic efficacy and cytotoxicity. • Ran in-vitro ADME studies to characterize compound pharmacokinetics, using HPLC for quantitative and qualitative analysis of biochemical compounds and fluorimetric glucose uptake and inhibition assays. • Operated and optimized 3D bioprinting to build in-vitro tissue models and organoids for disease modeling, owning experimental design, protocol optimization, and quality control documentation. • Maintained and cryopreserved mammalian cell lines at high viability to supply the bioink formulation process, and authored SOPs for routine laboratory procedures. • Built market research and competitive analysis for the tissue platform, and contributed to investor decks and grant proposals used in fundraising.	Aug 2023 – Aug 2024
Tranalab Pvt Ltd, Center for Human Genetics, Bachelor's Thesis, Bangalore, India • Designed and executed a complete expression and purification process for M4, a 35 kDa recombinant membrane protein domain, as a bachelor's thesis project ranked first in the cohort. • Identified solubilization as the yield-limiting step and engineered a modified buffer using non-ionic detergents, increasing soluble protein recovery by roughly 48% (A280 0.46 to 0.68). • Ran the upstream and downstream train end to end: expression in <i>E. coli</i> BL21(DE3) under IPTG induction, cell lysis by probe sonication, inclusion body solubilization, and Ni-NTA affinity chromatography (IMAC) with a stepwise urea wash gradient and imidazole elution. • Confirmed product identity by SDS-PAGE with Coomassie staining, quantified yield by A280, buffer-exchanged by overnight dialysis, and assessed refolding by sugar-affinity chromatography to verify retained binding function.	Jan 2023 – Aug 2023

Leadership and Projects

Vice President, Northeastern University, Graduate Biotech-Bioinfo Association, Boston, MA • Led executive board operations and organizational strategy, managing communications and cross-committee coordination while planning networking events, workshops, and industry panel discussions for member engagement.	Dec 2024 – Jan 2026
CurcuNano, Chief Executive Officer, BIOT 5219 Biotech Enterprise, Northeastern University • Led the venture team as CEO for a curcumin nanoformulation concept, developing the business case and market positioning across a semester-long project. • Owned the funding ask in the final investor pitch, covering raise amount, use of funds, and investment rationale, defended to a faculty panel.	Spring 2026